Changes in the physical, mechanical, chemical and energy parameters of coffee husk pellets during short-term storage

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Highlights

- Waste in the form of coffee husks is a good material for the production of pellets.

- Significant changes in pellet parameters occur during the first two weeks.

- High mechanical durability and unit density are achieved by pellets with MC < 20 %

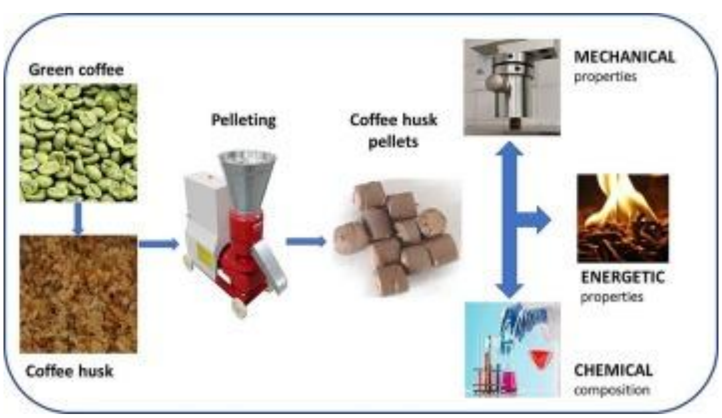
- The compressive strength of pellets is correlated with storage time and moisture loss.

- Coffee husks contain extractive substances and carbohydrates and can be used to produce bioenergy.

Abstract

Coffee husks are waste in the coffee roasting process, constituting about 12 % of the mass of the beans, generating a significant burden on the environment if they are not correctly disposed of. Pellets made from them can be sold or burned in the manufacturing plant. The justification for the undertaken research was to indicate the period after production of the agglomerate to obtain good technical and energetic quality. The research material in the form of unprocessed and pressure-agglomerated husks was obtained from a coffee roastery in Poland. The pellets were tested over 11 weeks, on the 3rd, 10th, 17th, 23rd, 52nd, and 78th day after their production. The aim of the study was to evaluate changes in parameters with significance for short-term storage (no longer than 2–3 months), transport and combustion. The changing moisture content of the pellet, which is more intense in the first days after its production, affects the physical and mechanical properties of the pellet produced; with time after its production, the unit and bulk density of the pellet increases. At the same time, intermolecular bonds become more robust, and the pellet is more complex and more resistant to external factors. The pellet acquires good properties after approx. two weeks from production, it is possible to transport and store it, and it becomes resistant to moisture absorption. The examined pellets were initially characterized by high moisture content (over 40 %, which declined during storage to approx. 7 %), an average bulk density of over 400 kg m^{-3}, a unit density of approximately 850 kg m^{-3}, and a mechanical durability of 96.5 %. As regards compressive strength, a decrease in the axial direction and an increase in the radial direction were observed in the tested pellets over time and with decreasing moisture content. In both cases compressive strain decreased. Dry pellets stored at a higher temperature (25 °C) and lower humidity (30 %) were found to absorb less moisture than those stored at a lower temperature (15 °C) but higher humidity (60 %). In turn, water absorption increased linearly with the time of immersion (from 5 to 30 min). The obtained elemental composition as well as a relatively high net calorific value (18.57 MJ kg^{-1}) indicate that coffee husk pellets are a suitable material for combustion and other applications. The chemical analysis of coffee husks showed a high proportion of extractive substances (36.89 %), which means that this waste material has the potential to be used.

Graphical abstract



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Introduction

The production and consumption of pellets are concentrated mainly in Europe and North America [1]. The imports of roasted and unroasted coffee to Poland in 2022 amounted to 196.8 thousand tons [2], representing a 1.9 % share in the world's imports, which makes Poland 14th among coffee-importing countries. In 2022, Poland saw a 4 % increase in coffee imports on the previous year, and, as compared to 2010, the amount of imported coffee increased by 61.5 thousand tons. Per capita annual coffee consumption in Poland is approx. 2.3 kg. The highest per capita annual coffee consumption is found among the Scandinavians: 12 kg among the Finns and 9.9 kg among the Norwegians. Such quantities of coffee generate large amounts of waste both during roasting as well as upon brewing and consumption. It is estimated that one ton of green coffee beans results in about 650 kg of spent coffee grounds (SCG) [3], [4], which translates into over 6.5 Mt of waste annually worldwide.

SCG and their uses have been discussed by many authors. SCG have been pelletized and combusted [5], torrefied [6], [7], converted into biodiesel and used as a mineral-rich fertilizer. Coffee husks are a by-product generated during coffee roasting. Depending on the roasting method (wet or dry) [8], they may contain the endocarp (the layer surrounding the seed) or be devoid of it. Husks account for about 12 % of coffee beans by weight, generating a significant environmental burden if disposed inappropriately. As reported by Manrique et al. [9], in Colombia this waste constitutes an energy source equivalent to over 49 PJ annually. Just like SCG, husks can also be combusted or pressure-agglomerated [10]. They may also serve as reinforcing additives to thermoplastic compounds [11], [12] and contain micro and nanocellulose. As reported by Abarca et al. [13], coffee waste is not a good source of fiber for food preparation, but it can be used as a component of pressure-agglomerated boards [8].

Some coffee roasting facilities sell husks in bulk or, following pressure agglomeration, as pellets or briquettes [10], [14]. The main drawbacks of husks, such as high moisture content, irregular shapes, and low bulk density (the high volume they require for storage), make it challenging to use them in unprocessed form in modern combustion systems. The disadvantage of low-density and large-volume materials is also high storage costs. Therefore, they need to be converted into denser fuels prior to combustion.

According to Seaenger et al. [15], one of the causes of the low utilization of coffee husks as a fuel, whether in raw or agglomerated form, is insufficient knowledge regarding their combustion characteristics. The physical and chemical properties of coffee husks and pellets produced from them, such as bulk density, ash melting temperature, elemental composition, and volatile content, determine the efficiency of the combustion process.

Such information would be useful in designing and operating combustion systems using coffee husk-based fuels.

Some of the physical properties of coffee husks [8], [15], [16] and their combustion or pelletization processes [9], [17] have already been discussed in the literature. However, the available studies mainly concern biomass materials mixed with various coffee additives with a view to obtaining and improving pellet durability and strength. The research results allowed us to determine the parameters of the agglomeration process (pressure, temperature, particle size) at which good-quality pellets are obtained. The use of additives to coffee husks improves the pellet's physical and mechanical properties. Additions of starch or molasses also improve the energy properties of the pellet, increasing its calorific value to about 30 MJ kg^{-1} [18]. According to the authors of various studies, additives used in the production of pellets from plant biomass, not only from coffee husks, have a positive effect on their physical, mechanical and energy parameters. As reported by de Souza et all [1] pellets produced with coffee residues can compete with wood pellets to generate heat. Few studies to date have been devoted to biomaterials composed of coffee husks alone.

Developing and supplementing knowledge about waste from the coffee roasting process and the parameters of the obtained agglomerate is essential. Moreover, there are no literature reports on changes in the physical and mechanical properties of coffee husk pellets during storage under varying humidity conditions. Therefore, the current study tested such pellets over several weeks after their production to assess alterations in their moisture content, density, mechanical durability, compressive strength, and water absorption. Additionally, the chemical composition, elemental composition, and calorific value of pellets were evaluated. The results can serve as information for coffee roastery, indicating how to handle waste in the form of husks and how to handle pellets in short-term storage. Despite many studies and benefits of using pellets made from various coffee wastes, it is advisable further to supplement knowledge about the properties of the agglomerate and possibly formulate recommendations for the industry.

The aim of the study was to determine the physical, mechanical, and chemical characteristics of fuel pellets produced from coffee husks and relate them to their transportation, storage, and energy properties. The paper describes the results of research conducted on pellets obtained from a Polish coffee roastery.

Study material

The studied coffee husks were generated as a waste product in the coffee roasting process. Immediately after being separated from the beans, they were pelletized in a 48 HPPA 200 machine (PROBAT-WERKE, Emmerich am Rhein, Germany) equipped with a die with a hole diameter of 8 mm. The maximum pelletizing capacity was 160 kg h^{-1}.

Coffee husks and pellets were collected at random directly from the production line of a coffee roastery in Poland. The materials were delivered to the laboratories of the

Moisture content, density, and mechanical durability of pellets

The relative moisture content of pellets decreased with the time of storage from 38.6 % on the third day after their production to 7.5–7.8 % as of day 52 (Table 1). ANOVA and Tukey's HSD *post-hoc* test revealed significant differences ($p < 0.05$) in mean values between all measurement days.

The pellets used in the study had a mean length of 9.83 ± 0.59 mm and a diameter of 10.02 ± 0.82 mm. Dry bulk density three days after pellet production was 367.2 ± 2.7 kg m^{-3}, and increased to 497.3 ± 4.6 kg m

Discussion

Coffee roasting facilities generate significant amounts of waste in the form of husks, which they may utilize internally, process, or sell. The composition of this waste is highly diverse due to the presence of a variety of organic compounds (such as lignin, cellulose, lipids, polysaccharides, and others), suggesting that this type of biomass may have multiple applications [50]. Cellulose is considered the main source of VOCs in biomass, while lignin serves as a natural binder in its structure

Conclusions

The aim of the present study was to investigate changes in the physico-mechanical, chemical, and energy parameters of coffee husk pellets associated with the variation in their moisture content during short-term storage.

Husks from the coffee roasting process can potentially serve as a substrate for the production of solid fuels (pellets), as confirmed by the obtained physicochemical results. They can also be utilized for the production of other fuels such as biodiesel, biogas, bioethanol, or

CRediT authorship contribution statement

Monika Aniszewska: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Arkadiusz Gendek:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Barbora Tamelová:** Writing – original draft, Resources,

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.